INCH-POUND

MIL-C-70845(AR)
AMENDMENT 3
30 January 1992
SUPERSEDING
AMENDMENT 2
12 April 1991

MILITARY SPECIFICATION

CARTRIDGE, 9MM, PRACTICE, TRACER, M939

This Amendment forms a part of MIL-C-70845 (AR), dated 25 July 1989, and is approved for use by U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1, under Standards, Military: Add "MIL-STD-1904 - Design and Test Requirements for Level A Ammunition Packaging"

PAGE 3

2.1.2, under Publications: Delete "TOP 1-2-601 - Laboratory Vibration Schedules, TECOM Test Operating Procedures" and "TOP 4-2-602 - Rough Handling, TECOM Test Operating Procedures".

PAGE 4

Delete paragraph "3.3.3" in its entirety and substitute:

"3.3.3 Bullet extraction. The average force required to separate the bullet from the cartridge case shall not be less than 40 pounds. The separation force for any individual cartridge shall not be less than 35 pounds."

Add paragraph "3.3.6.1" as follows:

"3.3.6.1 Alternate dispersion requirement. The average linear standard deviation, both horizontal and vertical, for all ten round targets fired at a range of 100 yards shall not exceed 3.9 inches."

3.3.8, line 2: Delete "± 1 mil" and substitute "± 1.8 mil".

PAGE 5

Delete paragraph "3.3.12" in its entirety.

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PAGE 6

Delete paragraph "3.5.2" in its entirety.

3.5.4, add to end of last sentence: "and the container shall be functional and suitable for its intended purpose."

Delete paragraph "3.5.5" in its entirety and substitute:

"3.5.5 Sequential rough handling. After the packaged cartridges are sequentially subjected to a 3 foot drop test, a loose cargo test and a 7 foot drop test, and bare cartridges are subjected to a 5 foot drop test, the container shall be functional and suitable for its intended function and the cartridges shall be safe to store, handle and fire. After packaged cartridges are subjected to a 40 foot drop test, the cartridges shall not detonate, deflagrate or exhibit propellant spillage and shall be safe to handle and dispose of."

PAGE 9

TABLE II., under PARAGRAPH REFERENCE/INSPECTION METHOD column:

Change "4.4.4.2" to "4.4.3.1".

TABLE II., under Examination for defects:

Change "Minor class 12 - 13" to "Minor class 10-11"

TABLE II., under NOTES:

Delete Note "2/" in its entirety.
Change "3/ See 4.4.4.2, Note 2." to "2/ See 4.4.4.2, Note 2."

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Table II., Delete "Ricochet 90 1/ 3.3.12 4.5.11" and
"Toxic Fumes 150 1/ 3.5.2 4.5.14"

PAGE 13

4.4.3.1, under Critical:

Delete "3. No primer 100% 3.1 Visual" and
"4. Cocked primer 100% 3.1 Visual" and
"5. Inverted primer 100% 3.1 Visual"

4.4.3.1, under Major: Delete "105. No radius on head (rim)" and substitute "105. No chamfer on head (rim)".

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4.4.3.1, under Major:

Add "116. No primer 3.1 Visual" and
"117. Cocked primer 3.1 Visual" and
"118. Inverted primer 3.1 Visual"

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Delete "216. Flat point (bullet) 3.1 Visual".

PAGE 17

4.4.4.2, under Pressure and velocity:

Change "+21°C 40" to "+21°C 30"
Change "-18°C 40" to "-18°C 30"
Change "+60°C 40" to "+60°C 30"

4.4.4.2, under Function and casualty:

Change "-18°C 150" to "-18°C 75"
Change "+60°C 150" to "+60°C 75"

PAGE 19

4.4.6, line 3: Delete "6.2" and substitute "6.3".

Delete paragraph "4.5.2.4" in its entirety and substitute:

"4.5.2.4 Cartridges with aluminum cases. The corrosion test specified in 4.5.12 shall be utilized as the residual stress test for aluminum cases."

Add new paragraph "4.5.2.5" as follows:

"4.5.2.5 Cartridges with cases of other materials. The test shall be conducted in accordance with procedures approved by the Government appropriate to the material(s) used to manufacture the cartridge."

4.5.4, line 2: Delete "MIL-STD-1168" and substitute "DOD-STD-1468".

4.5.5, line 4: Delete "MIL-STD-1168, section 4" and substitute "DOD-STD-1468, sections 4 and 12".

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4.5.7, line 2: Delete "MIL-STD-1168" and substitute "DOD-STD-1468".

Delete paragraph "4.5.11" in its entirety.

PAGE 21

4.5.12, line 2: Delete "After 48 hours" and substitute "After 24 hours".

Delete paragraph "4.5.14" in its entirety.

Delete paragraph "4.5.16" in its entirety and substitute:

"4.5.16 Secure cargo (transportation/vibration). Two wire-bound boxes (4000 rounds total) shall be subjected to the secured, non-tactical vibration test as specified in paragraph 6.3.1 of MIL-STD-1904. Testing shall be performed at ambient temperature only. Following testing, failure of the containers to comply with the criteria specified in paragraph 7.4 of MIL-STD-1904 shall be cause for rejection. The containers shall be unpacked and the cartridges visually examined for damage. Failure of the cartridges to comply with the criteria specified in paragraph 7.3 of MIL-STD-1904 shall be cause for rejection. All damage to containers and cartridges shall be recorded. The cartridges shall be repacked and subjected to the sequential rough handling test (4.5.17). The secure cargo test shall be performed by the Government."

Delete paragraph "4.5.17" in its entirety and substitute:

"4.5.17 Sequential rough handling. This test shall be performed after the secure cargo test (4.5.16) utilizing the same cartridges and packaging. One wire-bound box (2000 rounds) shall be conditioned at a temperature of 160 $\pm$ 5 degrees F and the second box shall be conditioned at -65 $\pm$ 5 degrees F. After conditioning for a minimum of 16 hours, both boxes shall be subjected to the following tests in the order specified:

a. Three foot drop test in accordance with paragraph 6.4 of MIL-STD-1904.

b. Loose cargo test in accordance with paragraph 6.13 of MIL-STD-1904 (two planes).

c. Seven foot drop test in accordance with paragraph 6.4 of MIL-STD-1904, with the exception that each box shall be dropped twice, once on the bottom and once on the box side.

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At the conclusion of the testing, failure of the containers to comply with the criteria specified in paragraph 7.4 of MIL-STD-1904 shall be cause for rejection. The containers shall be unpacked and the cartridges visually examined for damage. Failure of the cartridges to comply with the criteria specified in paragraph 7.3 of MIL-STD-1904 shall be cause for rejection. All damage to containers and cartridges shall be recorded. A random sample of 150 cartridges shall be selected from each of the two wire-bound boxes (300 rounds total). A sample of 50 cartridges from the 300 round sample shall be subjected to the following 5 foot drop test.

Each of the fifty rounds shall be dropped twice on the same surface utilized in the 7 foot drop test from a height of 5 feet ± 3 inches in the orientations specified below:

	Drop Orientation	
Round #	Drop #1	Drop #2
1 & 2	Horizontal	Horizontal
3 & 4	"	Base down
5 & 6	"	Nose down
7 & 8	"	45°, base down
9 & 10	"	45°, nose down
11 & 12	Base down	Horizontal
13 & 14	"	Base down
15 & 16	"	Nose down
17 & 18	"	45°, base down
19 & 20	"	45°, nose down
21 & 22	Nose down	Horizontal
23 & 24	"	Base down
25 & 26	"	Nose down
27 & 28	"	45°, base down
29 & 30	"	45°, nose down
31 & 32	45°, base down	Horizontal
33 & 34	"	Base down
35 & 36	"	Nose down
37 & 38	"	45°, base down
39 & 40	"	45°, nose down
41 & 42	45°, nose down	Horizontal
43 & 44	"	Base down

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45 & 46	"	Nose down
47 & 48	"	45°, base down
49 & 50	"	45°, nose down

At the conclusion of the drop testing, the 50 round sample shall be combined with the remaining 250 rounds of the original sample and shall be fired at ambient temperature in accordance with the procedures for function and casualty testing specified in 4.5.7. The reliability (total # of defects/total # of rounds fired) demonstrated in this test shall be no more than 10 per cent below the reliability demonstrated during the initial ambient temperature function and casualty test. The remaining metal container from each box shall be repacked and combined into one wire-bound box (the one in best condition) and shall be subjected to a 40 foot drop test at ambient temperature in accordance with paragraph 6.12 of MIL-STD-1904. Following this test the cartridges shall be safe to handle and dispose of as specified in 3.5.5. The container need not be functional. The test results shall be documented in accordance with paragraph 5.3 of MIL-STD-1904 and be supplemented with photographs documenting the procedures and damages. The sequential rough handling test shall be performed by the Government."

PAGE 22

4.5.18, line 3: Delete "by the contractor." and substitute "by the Government, if required."

PAGE 24

6.7, line 6: Delete "0.65" and substitute "1.5".

PAGE 28

60.2, line 4: Delete "Standard Deviation $= \frac{n \quad d^2 \quad - \left(\quad d \right)^2}{n \ (n-1)}$"

and substitute "Standard Deviation $= \sqrt{\frac{n(\Sigma d^2) - (\Sigma d)^2}{n(n-1)}}$ "

PAGE 30

40.1, first sentence: Delete "Trace performance shall be made at weapon and at points 65 meters and 450 meters beyond the muzzle of the gun" and substitute "Observation of trace performance during night testing shall be made at the training weapon and at points 65 meters and 450 meters beyond the muzzle of the trainer".

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40.2, Delete "Observation for day testing." and substitute "Observation at the trainer."

Add paragraph "40.5" as follows:

"40.5 Observation for day testing. Observation of trace performance during day testing shall be made from directly behind the trainer only."

PAGE 31

50.1.1, line 3: Delete "The trace testing shall be performed at night." and substitute "One half of the sample cartridges shall be tested at night. The remaining cartridges shall be tested during the day."

50.1.2, Delete "50.1.2" in its entirety and substitute:

"50.1.2 Test weapon. All trace testing shall be performed in the M287 Sub-caliber training weapon. The trainer shall be at ambient temperature prior to firing each test."

PAGE 33

Delete paragraph "40.1.3" in its entirety and substitute:

"40.1.3 Noise level measurement. A noise level measurement device shall be positioned such that the noise level is measured at the right handed gunner's head position and at a point 1 meter to the right of the trainer."

Delete paragraph "40.1.4" in its entirety.

40.2.2, Add at end of paragraph: "One half of the sample rounds shall be fired for noise level measurement at each of the specified positions."

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